Career Path Generator — Feature Overview

Core System

- **User Profile Management —** Stores each user's core skills, interest domain, experience level, and leadership score. Acts as the foundation for all analysis and recommendations the system makes.
- **Career Alignment Engine —** Analyzes the full user base and classifies profiles into High, Moderate, or Low alignment categories based on how well their current skills match market demand. Gives a bird's eye view of where people stand.
- **Top Emerging Career Clusters —** Maintains a live-updated table of the most in-demand career domains (like AI & Data Engineering, Cybersecurity, Product Management) with growth trends and demand scores sourced from market data.

Intelligence Features

- **Skill Gap Radar Chart —** Visually shows the delta between a user's current skillset and the requirements of their target role. Displays a readiness percentage so users know exactly what's missing and how close they are.
- **Interactive Career Roadmap Visualizer —** A node-based visual map where each node represents a career role and edges show the transition path. Each node is clickable and reveals time estimates, required skills, salary jump, and risk level for that transition.
- **Scenario-Based Transition Paths —** Goes beyond just suggesting roles. Considers life stage factors like burnout, purpose shift, stress tolerance, and family changes to recommend humane, realistic transition paths like Engineer → Technical Educator → EdTech Leadership.

AI-Powered Features

- **Resume Upload → Auto Profile Fill (VLM) —** User uploads their resume as a PDF or even a photo, and a Vision Language Model automatically extracts skills, experience, and domains to populate their profile. Eliminates all manual data entry.
- **"Ask Your Profile" Chat (Claude API) —** A conversational interface where users can ask natural language questions about their own career data. Questions like "What roles can I reach in 2 years?" or "If I learn Docker, how does my alignment change?" get answered in real time using their profile as context.

Analytics & Ethics Features

- **Cohort Benchmarking —** Compares a user's profile against peers with similar backgrounds in the dataset. Shows percentile rankings for leadership score, skill breadth, and alignment — so users understand where they stand relative to others on the same path.
- **Bias Audit Panel** — Since the system explicitly positions itself as Ethical & Bias-Aware, this panel surfaces potential biases in the recommendations. Flags things like geography bias, gender-neutral scoring gaps, or over-weighting of leadership scores for career changers. Makes the AI accountable, not just compliant.

Reporting & Dashboard

- **Career Alignment Overview Dashboard —** High-level summary view showing total profiles analyzed, alignment distribution, and cluster trends. The command center for understanding the full dataset at a glance.
- **Market Trend Analysis —** A dedicated report that tracks how demand scores for different clusters shift over time, helping users and advisors make forward-looking decisions rather than reacting to outdated data.